# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** The value of $101^3$ is: 1

**(a)** 1030300 **(b)** 1030301 **(c)** 1030303 **(d)** 1030302

**2.** If the distance between $(x, 2)$ and $(1, 6)$ is $5$ and $x > 1$, then $x =$ 1

**(a)** 3 **(b)** 6 **(c)** 4 **(d)** 5

**3.** If $x = 4$ is the zero of $kx - 28$, then $k =$ 1

**(a)** 7 **(b)** 8 **(c)** 6 **(d)** 9

**4.** $1.\overline{27}$ in the form $\frac{p}{q}$ is: 1

**(a)** $\frac{7}{6}$ **(b)** $\frac{13}{11}$ **(c)** $\frac{14}{11}$ **(d)** $\frac{15}{11}$

**5.** If $a + b + c = 0$, then $a^3 + b^3 + c^3$ for $a = 2$, $b = 3$, $c = -5$ equals: 1

**(a)** -90 **(b)** -91 **(c)** -89 **(d)** -88

**6.** The fourth vertex of the parallelogram with consecutive vertices $(1, 1)$, $(4, 2)$, $(6, 5)$ (in order) is: 1

**(a)** $(-3, 4)$ **(b)** $(3, 4)$ **(c)** $(3, -4)$ **(d)** $(4, 3)$

**7.** If a linear polynomial $ax + b$ has zero $0$, then: 1

**(a)** $a = b$ **(b)** $b = 0$ **(c)** $b = 1$ **(d)** $a = 0$

**8.** The value of $0.\overline{9}$ is: 1

**(a)** $0.9$ **(b)** $1$

**(c)** $0.99$ **(d)** just less than $1$

**9.** If $a + b + c = 10$ and $a^2 + b^2 + c^2 = 38$, then $ab + bc + ca =$ 1

**(a)** 30 **(b)** 31 **(c)** 33 **(d)** 32

**10.** A point $P(x, 0)$ on the $x$-axis is $10$ units from $(-3, 8)$ with $x > -3$. The value of $x$ is: 1

**(a)** 4 **(b)** 5 **(c)** 2 **(d)** 3

**11.** For the pattern with $n$th term $6n - 1$, the sum of the 1st and 10th terms is: 1

**(a)** 66 **(b)** 63 **(c)** 65 **(d)** 64

**12.** The length of the repeating block of $\frac{1}{41}$ is: 1

**(a)** 5 **(b)** 4 **(c)** 7 **(d)** 6

**13.** If $x^2 + \frac{1}{x^2} = 14$, then $\left(x + \frac{1}{x}\right)^2 =$ 1

**(a)** 17 **(b)** 18 **(c)** 15 **(d)** 16

**14.** The points $(a, 0)$, $(0, a)$ and $(0, 0)$ form a triangle which is: 1

**(a)** scalene **(b)** obtuse-angled

**(c)** isosceles right-angled **(d)** equilateral

**15.** For which value of $n$ is $13n - 4 = 217$? 1

**(a)** 17 **(b)** 16 **(c)** 18 **(d)** 19

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** If $p(x) = 3x + 4$, find $p(p(x))$ and its value at $x = 2$. 2

**17.** Find the smallest natural number $n$ such that $\frac{n}{420}$ has a terminating decimal expansion. 2

**18.** Find the value of $103^3$ using an identity. 2

**19.** Find the distance between the points $(-7, 4)$ and $(2, -8)$ and its square root form. 2

**20.** Find the area of the triangle formed with the axes by the line $y = 2x + 14$. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** If $a + b + c = 0$, show that $a^3 + b^3 + c^3 = 3abc$ for $a = 1$, $b = 4$, $c = -5$. 3

**22.** Find the length of the median $AD$ of the triangle with $A(0, 0)$, $B(42, 0)$, $C(0, 56)$, where $D$ is the midpoint of $BC$. 3

**23.** The $n$th term of a pattern is $14n + 3$. Find the number of terms that are less than 143 and the sum of the first two terms. 3

**24.** Given that $\sqrt{2}$ is irrational, prove that $3 + \sqrt{2}$ is irrational. 3

OR

Factorise $8x^3 - 27$. 3

**25.** Find the points on the $x$-axis whose distance from $A(-1, 24)$ is $26$ units. 3

OR

Find the point of intersection of $y = 7x - 3$ and $y = 3x + 9$. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** The point $P(x, 0)$ is equidistant from $A(3, 4)$ and $B(9, -4)$. Find $P$ and the distance $PA$. 5

OR

A line passes through $(1, 15)$ and $(4, 24)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

**27.** Simplify $(\sqrt{7} + \sqrt{5})^2 + (\sqrt{7} - \sqrt{5})^2$ and $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$. Which are rational? 5

OR

If $a + b + c = 6$ and $ab + bc + ca = 11$, find $a^2 + b^2 + c^2$ and, if $abc = 6$, evaluate $a^3 + b^3 + c^3$. 5

**END OF QUESTION PAPER** — **Total: 50 Marks**